Magnetic Resonance Imaging

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Lärandemål 1/3

1. Vad som mäts med MR
2. Varför väte går att avbilda med MR och varför just väte är ett lämpligt ämne att avbilda
3. Vad som händer när en atomkärna med spinn placeras i ett starkt magnetfält och varför
4. Följderna av en förändring i det externa magnetfältet B0
5. Ingående delar i en MR-utrustning samt deras funktion
6. RF-pulsens roll för generering av MR-signal

Lärandemål 2/3

7. MR-signalens uppkomst och utseende samt hur den detekteras
8. Hur kontrast skapas i MR-bilder
9. Hur parameterval i MR-utrustningen samt vävnadsegenskaper påverkar kontrasten i bilden
10. Hur MR-signalen lokaliseras i x, y – och z-led
11. Hur bildkvaliteten kan förbättras i en MR-bild
12. Vad som bestämmer temporal och spatiell upplösning i en MR-bild

Lärandemål 3/3

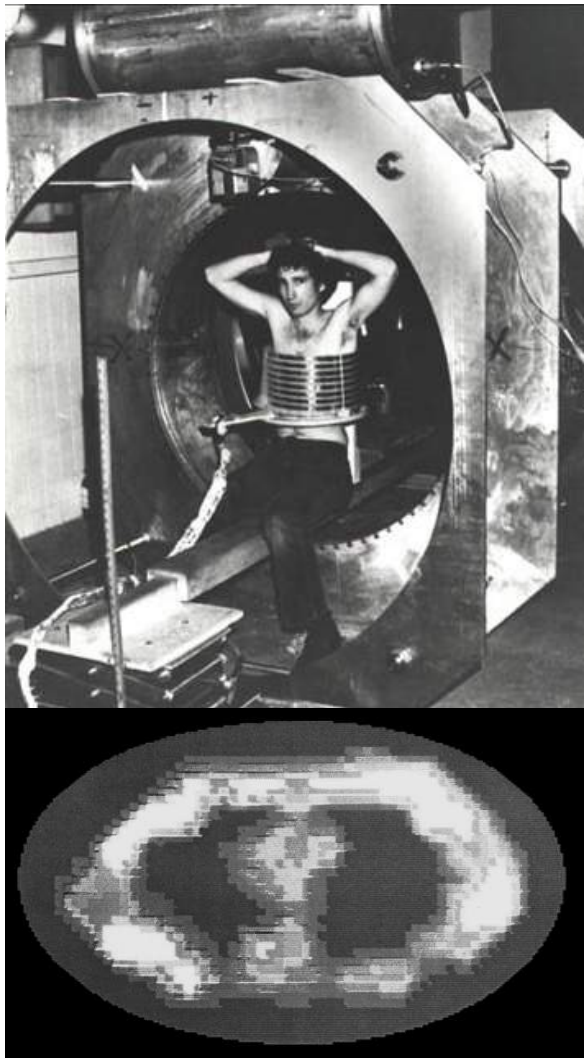
13. Hur spin-echosekvensen fungerar samt översiktligt känna till att det finns andra pulssekvenser
14. Säkerhetaspekter kopplat till MR-avbildning
15. Den kliniska nyttan av MR-avbildning samt redogöra för vilka kliniska situationer som MR är en lämplig avbildningsmetod
16. Vilka artefakter som kan uppstå vid MR-avbildning och hur de uppstår
17. MR-teknikens styrkor och svagheter jämförelse med andra tekniker?

Part I

Introduction

- Non-invasive
- No ionizing radiation
  - Low risk
  - Scan and rescan patients for follow-up
  - Scan healthy volunteers
- Excellent soft-tissue contrast
  - Not attenuation based (like x-ray)
- Versatile
  - Measure blood flow, diffusion, perfusion, cardiac motion, molecular content, metabolism
  - Most technological advances are now based on software

MRI then...



0.07T, Lund 1983



Philips 1.5T, 1985



...and now!



Brain at 100 micron resolution

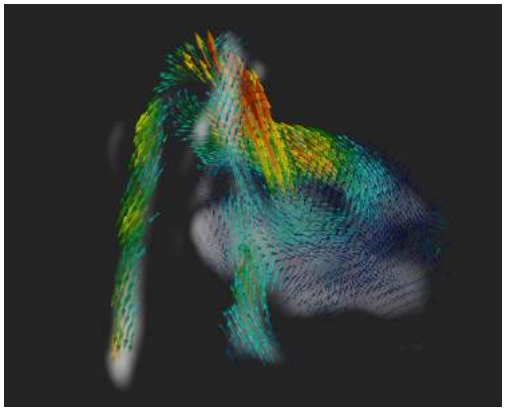
Siemens FREE.Max 0.55T
(University of Michigan 2021)



Siemens Terra.X 7T (2022)



In-utero cardiac 4D flow



Hyperfine Swoop 64mT

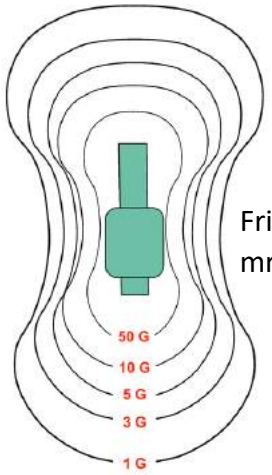


Source	Approximate Magnetic Field Strength
Neuron depolarization (imaged by MEG)	0.5 pT (5 x 10^{-13} T)
Earth's magnetic field	0.5 G (50 μT)
Refrigerator magnet	50 G (5 mT)
Junkyard electromagnet	1 T
Clinical MRI scanners	0.5 - 3.0 T (typical)
Research MRI scanners (human)	7.0 T – 11.7 T
Laboratory NMR spectrometers	6 - 23 T
Largest pulsed field created in lab nondestructively	97 T
Largest pulsed field created in lab (destroying equipment but not the lab)	730 T

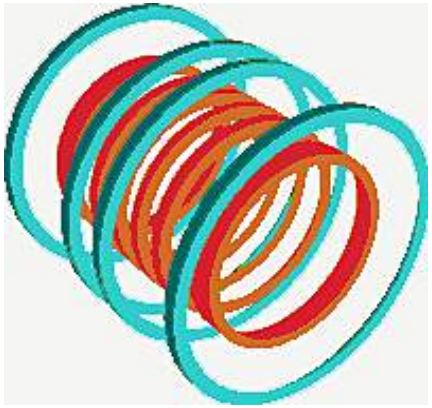
MRI Magnet design

- Niobium–titanium (Nb-Ti) multifilaments in copper matrix
- Super conducting at $T_c < 10$ K and $H_c < 15$ T
- 40 km wire, 500 A current for 1.5 T
- Helium bath
  - Capacity 1400 L, filled to 800 L
  - Boiling point 4.2 K
- No losses due to current, but due to cosmic radiation
  - Boiled off helium is recondensated using a cooling head for “zero boil-off”

Fringe fields

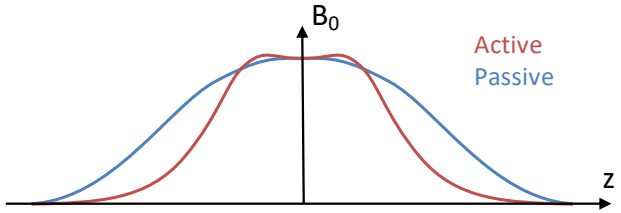


Fringe field (1 G = 0.1 mT)
mri-q.com

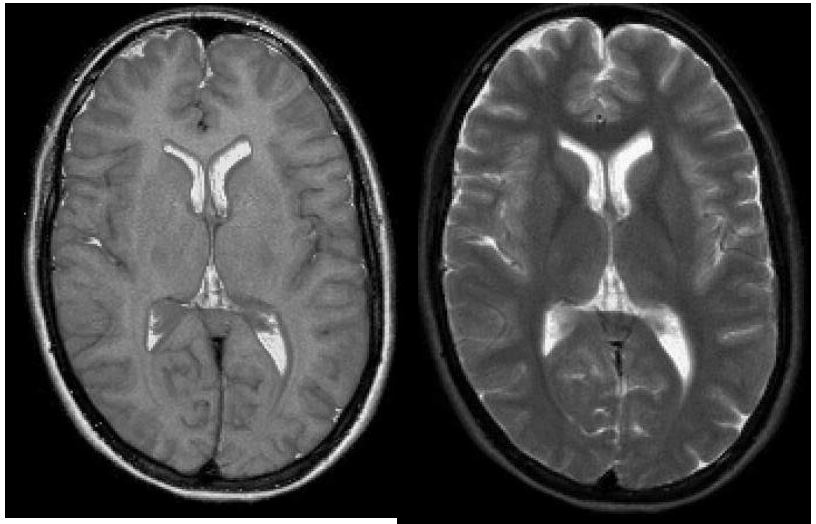


Active shield coils (blue) are in series with the main magnet windings (orange) but carry current in the opposite direction.
mri-q.com

- Actively shielded
  - Opposing field generated in series
  - Reduces fringe field
  - Higher spatial gradients (attractive force)

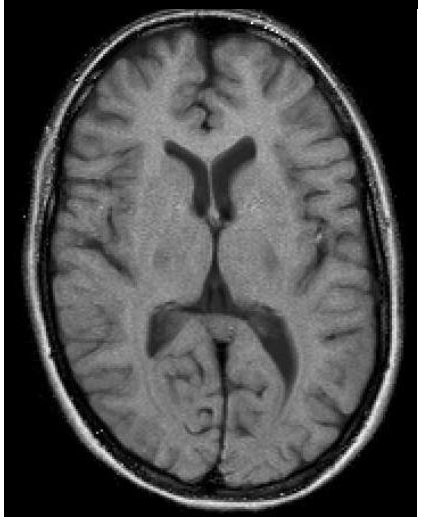






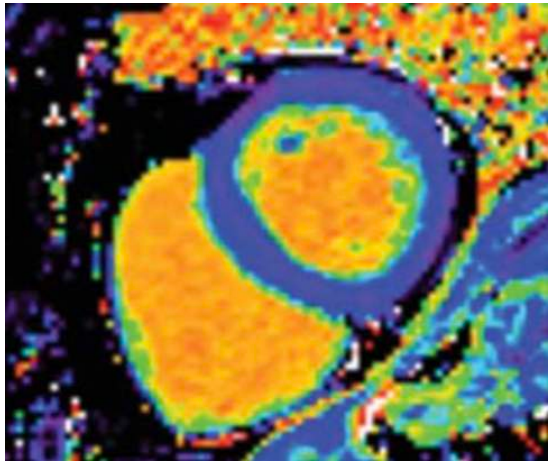




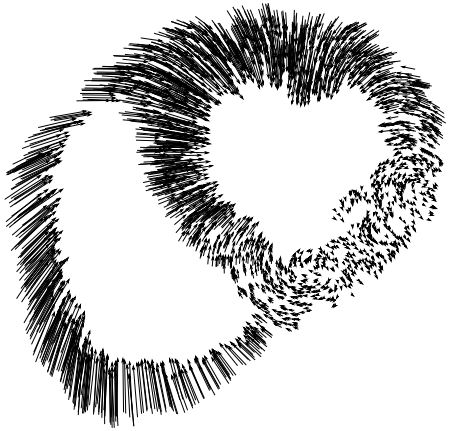




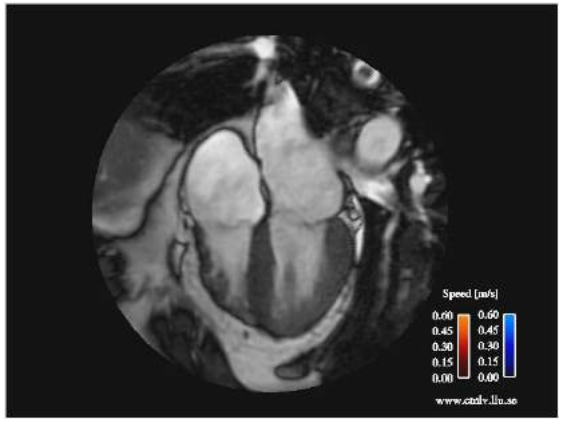
Diffusion tensor imaging



Myocardial tissue characterization



Myocardial motion



Blood flow

Nobel Prize winners



Rabi

- 1945: Isidor Isaac Rabi
  - For his resonance method for recording the magnetic properties of atomic nuclei
- 1952: Felix Bloch and Edward Purcell
  - for their development of new methods for nuclear magnetic precision measurements and discoveries in connection therewith
- 2003: Peter Mansfield and Paul C. Lauterbur
  - for their discoveries concerning magnetic resonance imaging



Bloch and Purcell



Mansfield and Lauterbur

Spin

1.2.5 The Pauli principle

The spin of particles has profound consequences. The *Pauli principle*[1] states:

two fermions may not have identical quantum states.

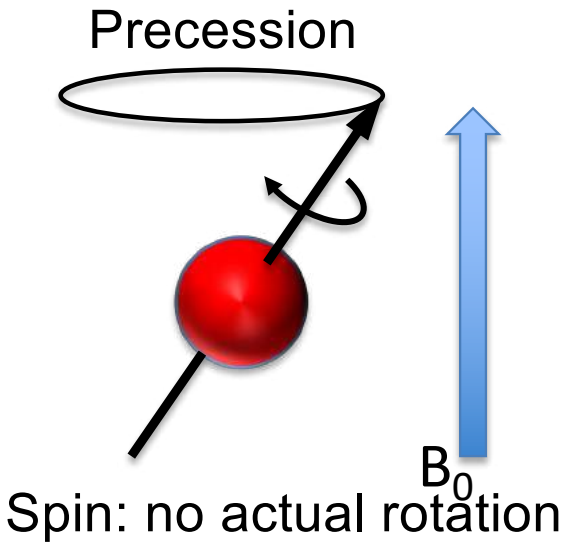
Since the electron is a fermion, this has major consequences for atomic and molecular structure. For example, the periodic system, the stability of the chemical bond, and the conductivity of metals, may all be explained by allowing electrons to fill up available quantum states, at each stage pairing up electrons with opposite spin before proceeding to the next level. This is called the *Aufbau principle* of matter, and is explained in standard textbooks on atomic and molecular structure (see *Further Reading*).

The everyday fact that one's body does not collapse spontaneously into a black hole depends on the spin-1/2 of the electron.

Nuclear magnetic moment and precession

- Atomic nuclei with odd mass number or odd charge number have non-zero spin
  - Magnetic moment, angular momentum
- The magnetic moment precesses in the presence of a magnetic field





The name of the game

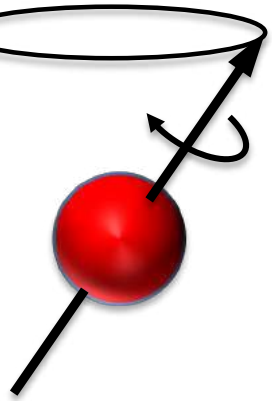
Larmor frequency

- Precession frequency proportional to spin magnetic moment and magnetic field

$$f_0 = \gamma B_0$$

γ Gyromagnetic ratio

B_0 External magnetic field



^{1}H at B_0=1.5T:

$$\gamma = 42.58 \text{ MHz / T}$$

$$f_0 = 64 \text{ MHz}$$

Quantization misconception

- Not only two possible states
  - Superposition of eigenstates: $\psi(t) = a(t)\psi_a + b(t)\psi_b$
- Quantization only occurs when measuring a single spin
- MRI always consider many spins, and can keep the continuous, non-quantized representation.

JOURNAL OF APPLIED PHYSICS VOLUME 28, NUMBER 1 JANUARY, 1957

Geometrical Representation of the Schrödinger Equation for Solving Maser Problems

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(Received September 18, 1956)

beam, such that the wave function for any one individual

$$\psi(t)=a(t)\psi_a+b(t)\psi_b \qquad (1)$$

dur… o
eig… e
single system corresponding to the energies $W+\hbar\omega_0/2$ and $W-\hbar\omega_0/2$ respectively. W is the mean energy of the two levels determined by velocities and internal interactions which remain unchanged. W will be taken as the zero of energy for each system. ω_0 is the resonant angular frequency associated with a transition between the two levels and is always taken positive.

It is usual to solve Schrödinger's equation with some perturbation V for the complex coefficients $a(t)$ and $b(t)$, and from them calculate the physical properties of the system. However, the mathematics is not always transparent and the complex coefficients do not give directly the values of real physical observables. Neither is it sufficient to know only the real magnitudes of a and b, i.e., the level populations and transition probabilities, when coherent processes are involved. We propose instead to take advantage of the fact that the phase of $\psi(t)$ has no influence so that only three real numbers are needed to completely specify $\psi(t)$. We constr… hich
have … 3-
vector…

$$r_1\equiv ab^*+ba^*$$
$$r_2\equiv i(ab^*-ba^*) \qquad (2)$$
$$r_3\equiv aa^*-bb^*.$$

(*) al… ime
depen… er's
equation which gives

$$i\hbar da/dt=a[(\hbar\omega_0/2)+V_{aa}]+bV_{ab} \qquad (3)$$

and similar equations for db/dt, da^*/dt, db^*/dt. The subscripts on V indicate the usual matrix elements. $V_{aa}=V_{bb}=0$ for most all cases of interest, and whenever these can be neglected compared to $\hbar\omega_0/2$, V need be neither small nor of short duration for the results to be exact. Using Eqs. (3) to find the differential equation f…

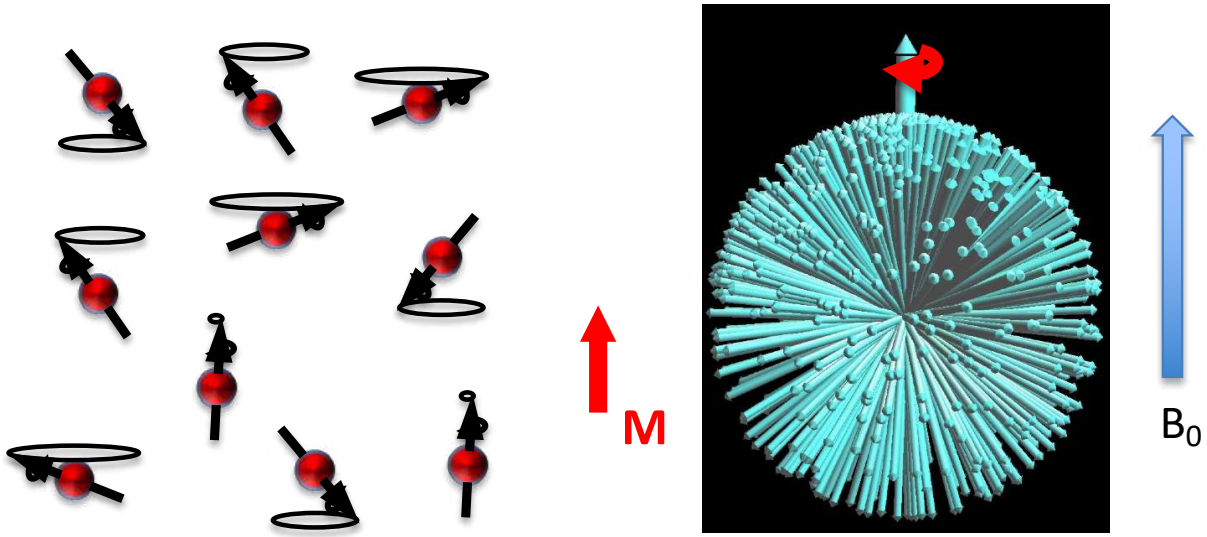
$$d\mathbf{r}/dt=\boldsymbol{\omega}\times\mathbf{r} \qquad (4)$$

where $\boldsymbol{\omega}$ is … ed by the three real components:

$$\omega_1\equiv(V_{ab}+V_{ba})/\hbar$$
$$\omega_2\equiv i(V_{ab}-V_{ba})/\hbar \qquad (5)$$
$$\omega_3\equiv\omega_0.$$

Net magnetization

- Many spins result in a net magnetization; M

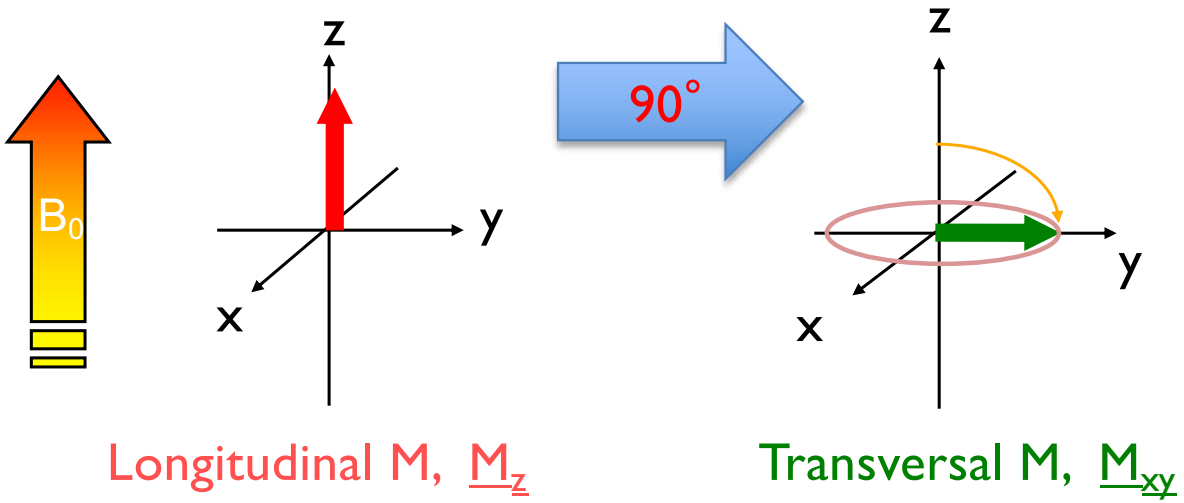


Hanson L, Concepts Mag Reson, 2008

Typical conditions (^{1}H at B_0 = 1.5 T, 310.15 K):
polarization ~ 0.00001 (100 ppm)

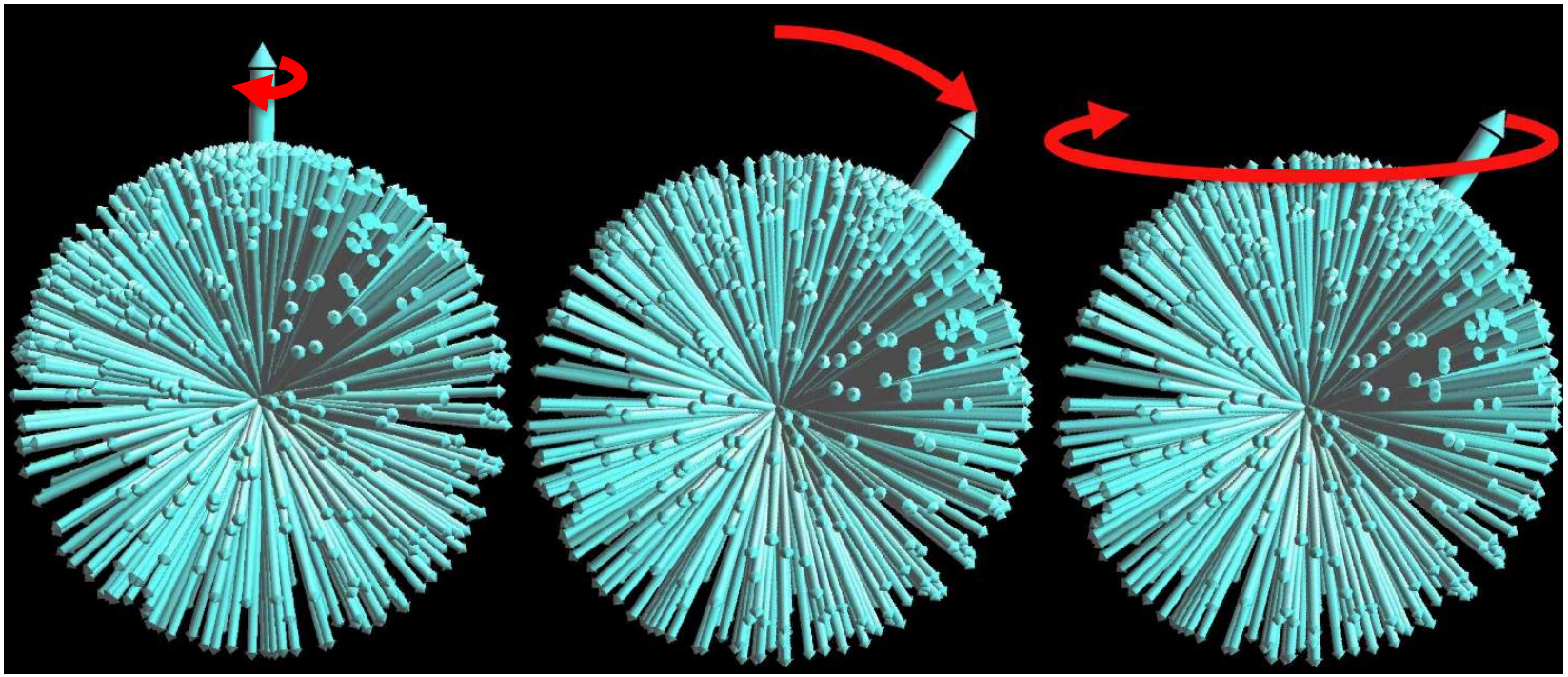
RF Excitation

- Time varying magnetic field, with Larmor frequency
  - 50-150 MHz, “radio frequency band”
  - Not propagating radio wave
- Turns magnetization away from the z axis



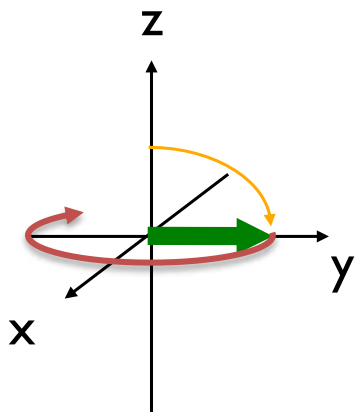
RF excitation and precession

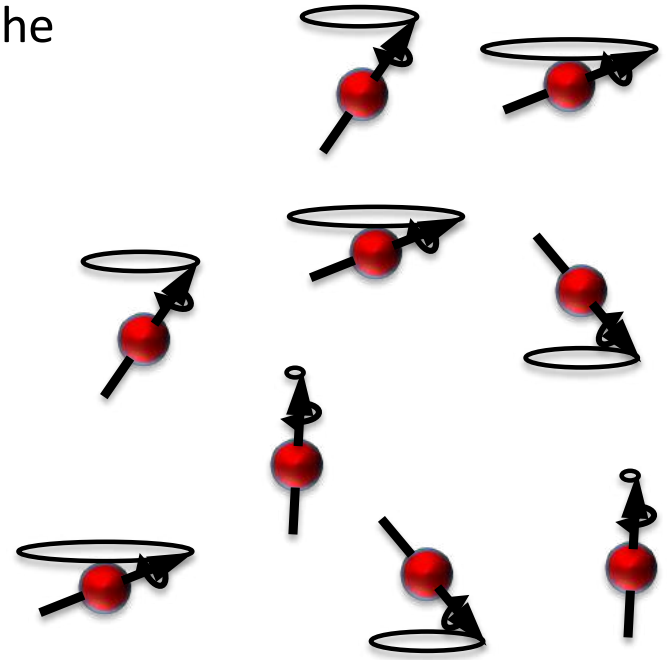
- RF excitation can be used to rotate magnetic moments
  - Net magnetization rotates and precesses



Net magnetization precession

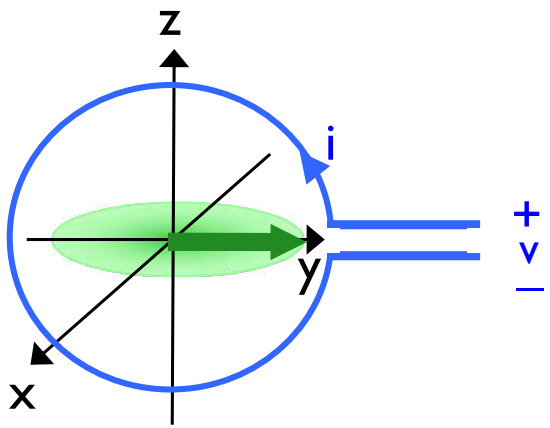
- After excitation, the net magnetization also precesses around the external magnetic field
  - Precession frequency is the Larmor frequency

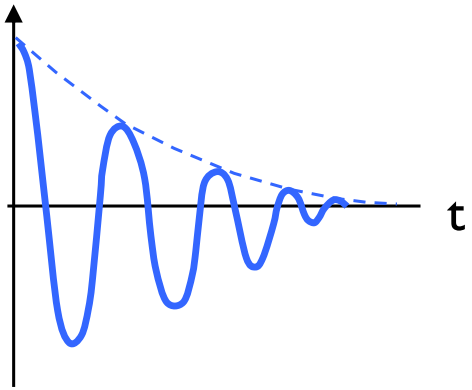




Signal reception

- Time varying current -> creates a magnetic field
- Time varying magnetic field -> induces a current
- Rotating M_{xy} induces a current in a conducting loop



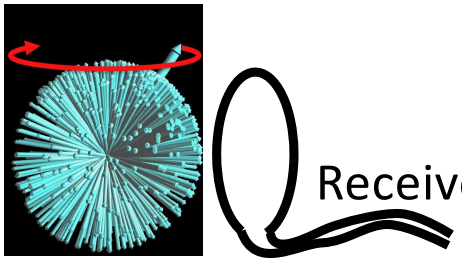


Signal reception

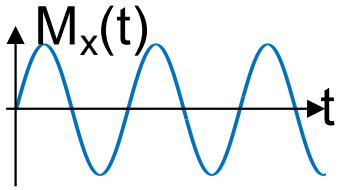
- Faraday's law of induction

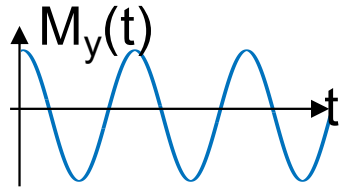
$$V(t) = -\frac{\partial \Phi}{\partial t}$$

Φ - Magnetic flux through coil
$V(t)$ - Induced voltage in coil



Receiver coil

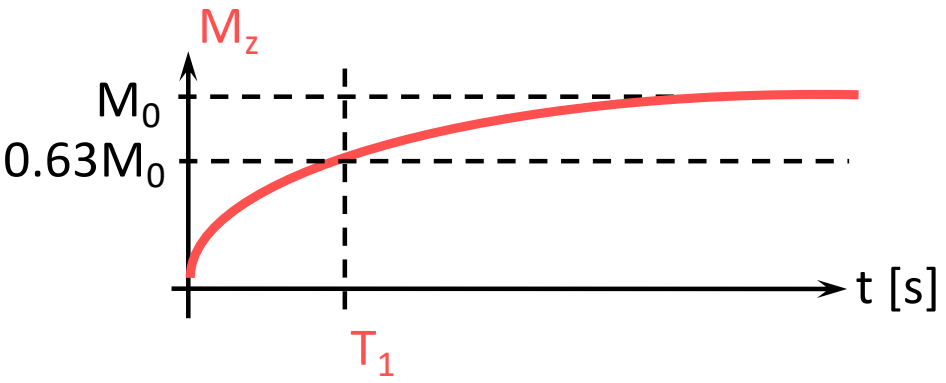




- Rotating M in xy plane induces a current in a conducting loop
- Signal is complex valued – proportional to $M_x + iM_y$

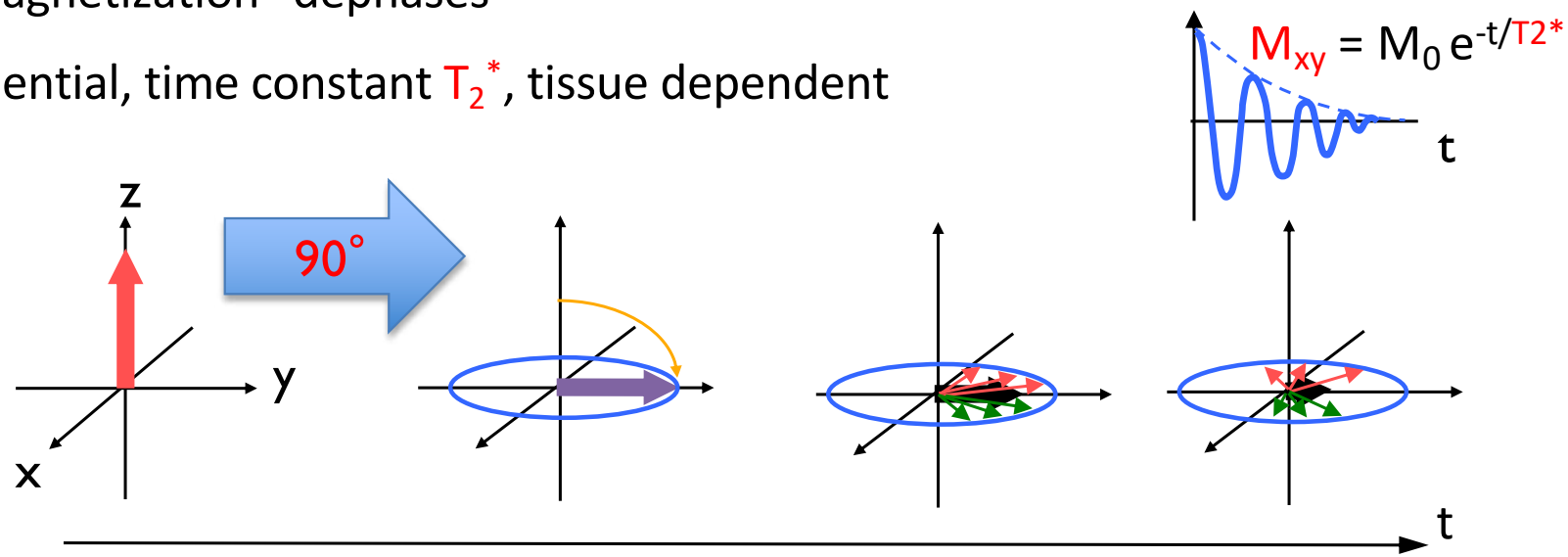
T_1 relaxation

- After excitation, the system returns to equilibrium
- M_z returns exponentially
  - $M_z = M_0 (1-e^{-t/T1})$
- Time constant: T_1
- Tissue dependent, typically in the order of 1 s for blood at 1.5 T



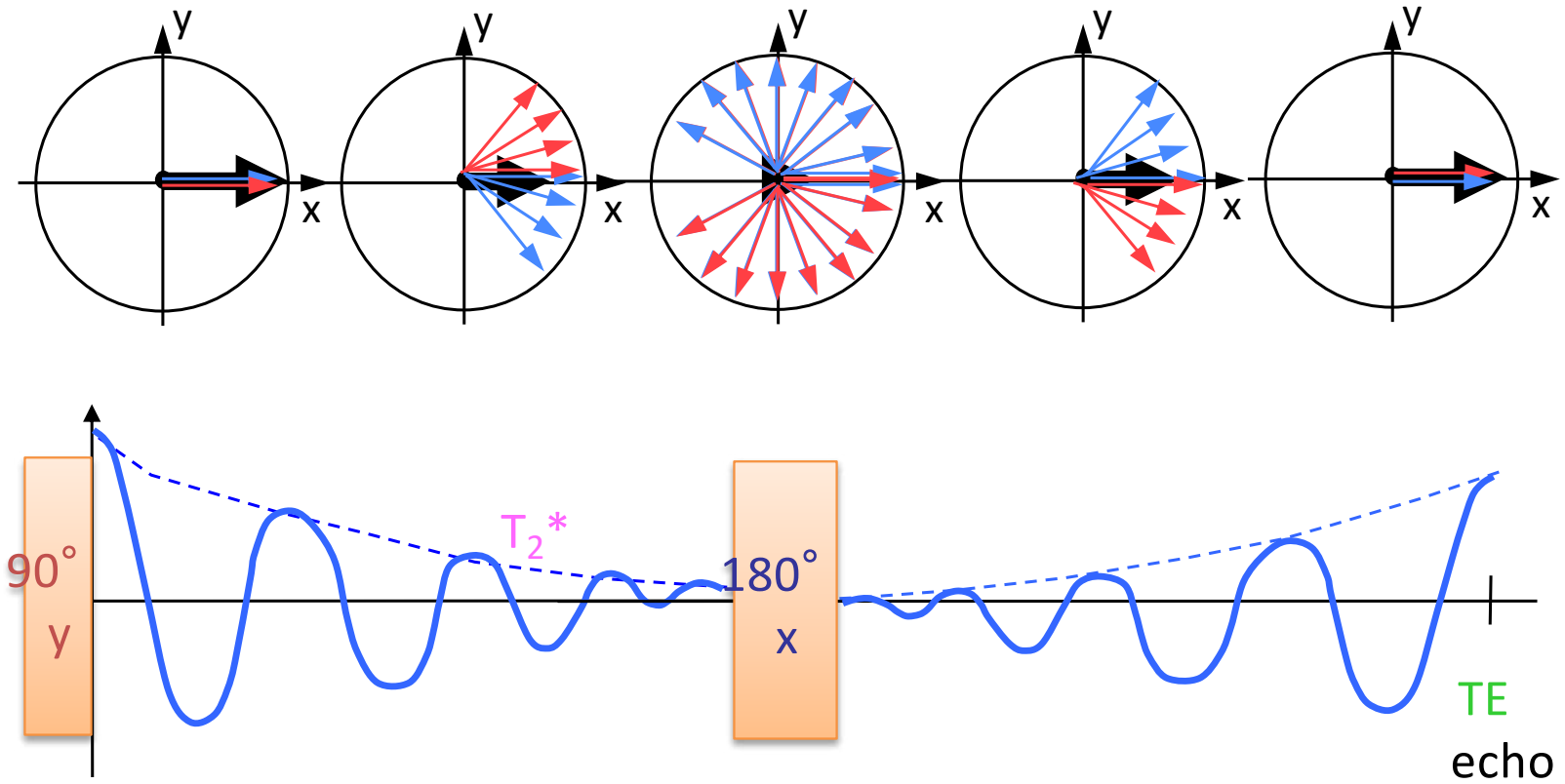
T_2^* relaxation

- Local variations of the magnetic field
  - Small magnetic field from surrounding electrons and nuclei
  - Non-homogeneous external magnetic field
- Varying magnetic field -> varying Larmor (precession) frequency
- Net magnetization “dephases”
- Exponential, time constant T_2^*, tissue dependent



The spin-echo pulse sequence

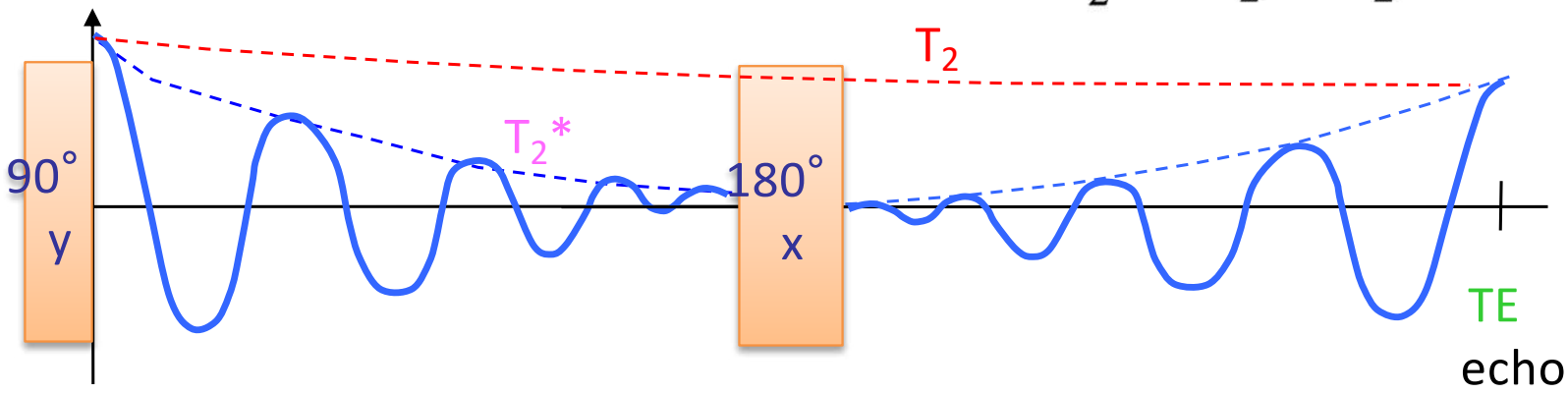
- Dephasing due to time-constant local variations can be refocused using an 180° pulse



T_2 and T_2^* relaxation

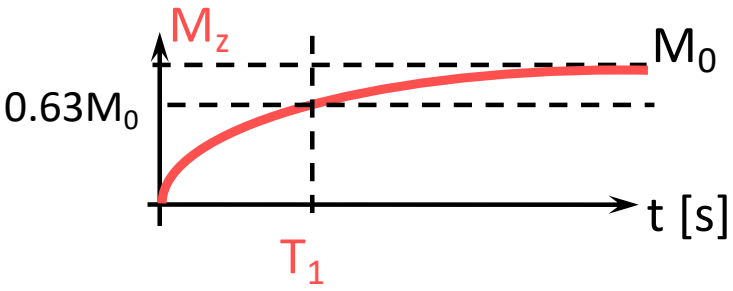
- Time-constant magnetic field variations can be refocused
  - Magnetic field inhomogeneities
- Time-varying variations cannot
  - Small magnetic field from surrounding electrons and nuclei
- Spin-echo pulse sequence not always applicable (no refocusing possible)
  - T_2^* is *apparent* T_2

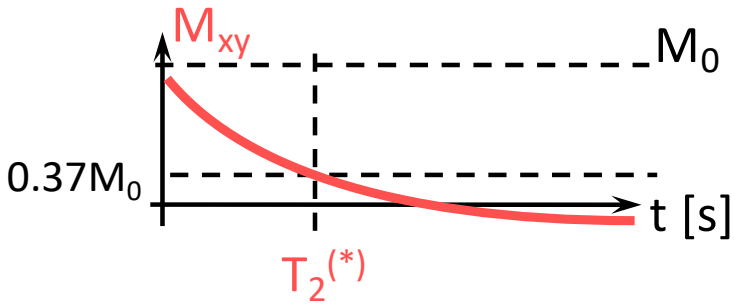
$$\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T_2'}$$



Relaxation summary

- T_1: Longitudinal relaxation
  - Recovery of M_z to equilibrium
- T_2: Transverse relaxation
  - Dynamic magnetic field variations
  - Reduction of M_{xy} magnitude due to dephasing
  - Unavoidable
- T_2^*: Apparent transverse relaxation
  - Static *and* Dynamic magnetic field variations
  - If static variations are note refocused, it manifests as apparent T_2
- $T_2^* < T_2 < T_1$





T_1 and T_2 values at 1.5 T

Tissue type	T_1	T_2
CSF	2500 ms	1000 ms
White brain matter	780 ms	90 ms
Gray brain matter	920 ms	100 ms
Blood	1200 ms	360 ms
Myocardium	880 ms	75 ms
Fat	260 ms	110 ms

Spin-echo pulse sequence

TE: Echo time
TR: Repetition time

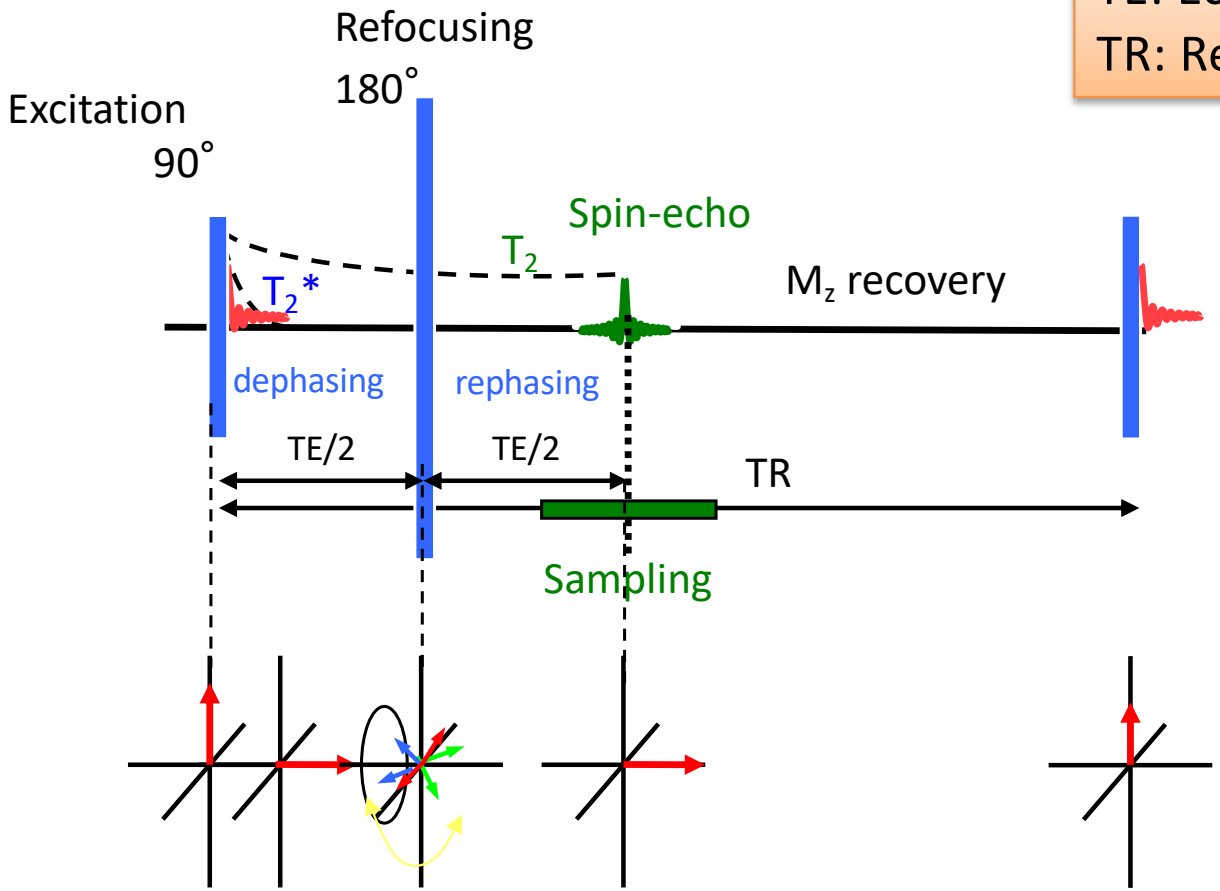
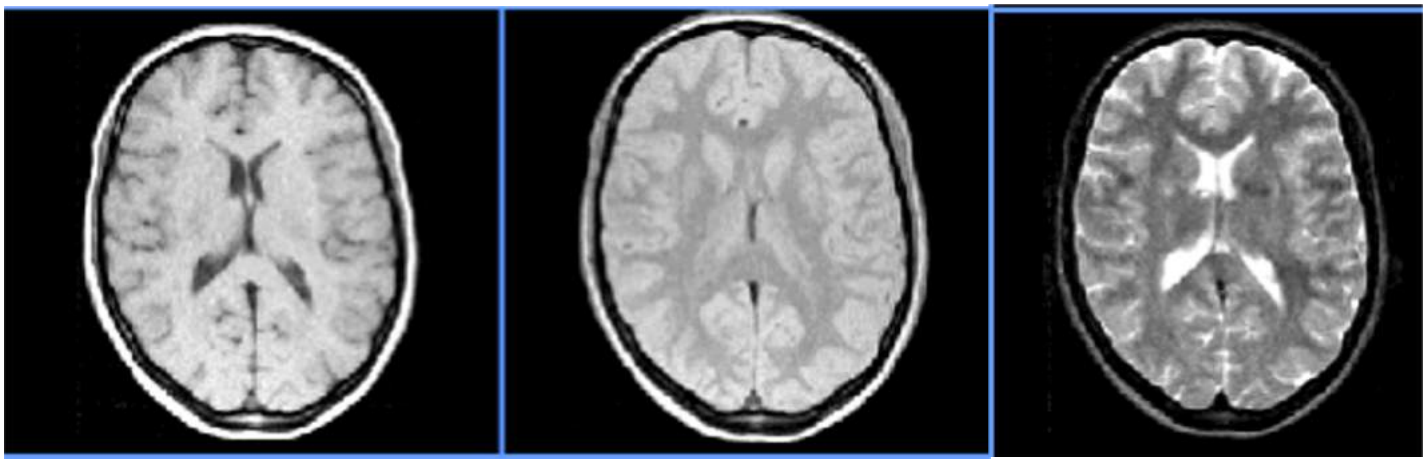
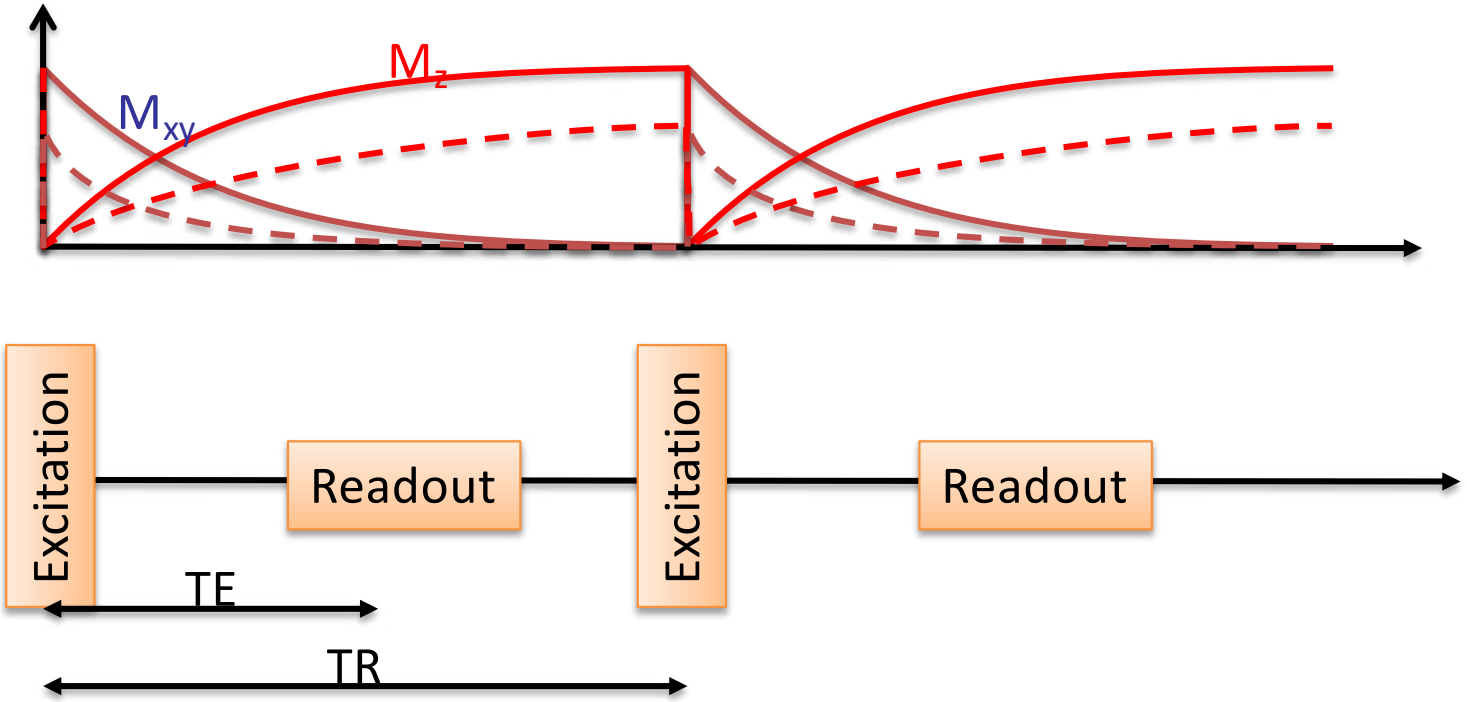


Image contrast

- By modifying pulse sequence timing (TE and TR), different image contrasts can be obtained
  - Tissue dependent T_1
  - Tissue dependent T_2

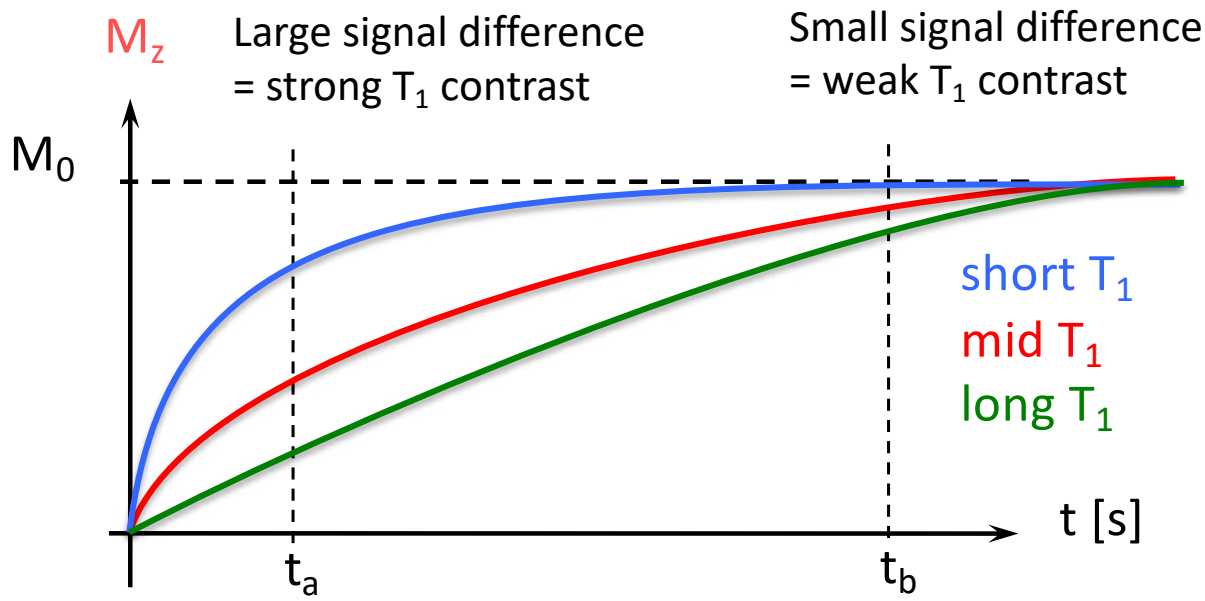




By adjusting TE and TR, signal magnitude will vary depending on T_1 and T_2

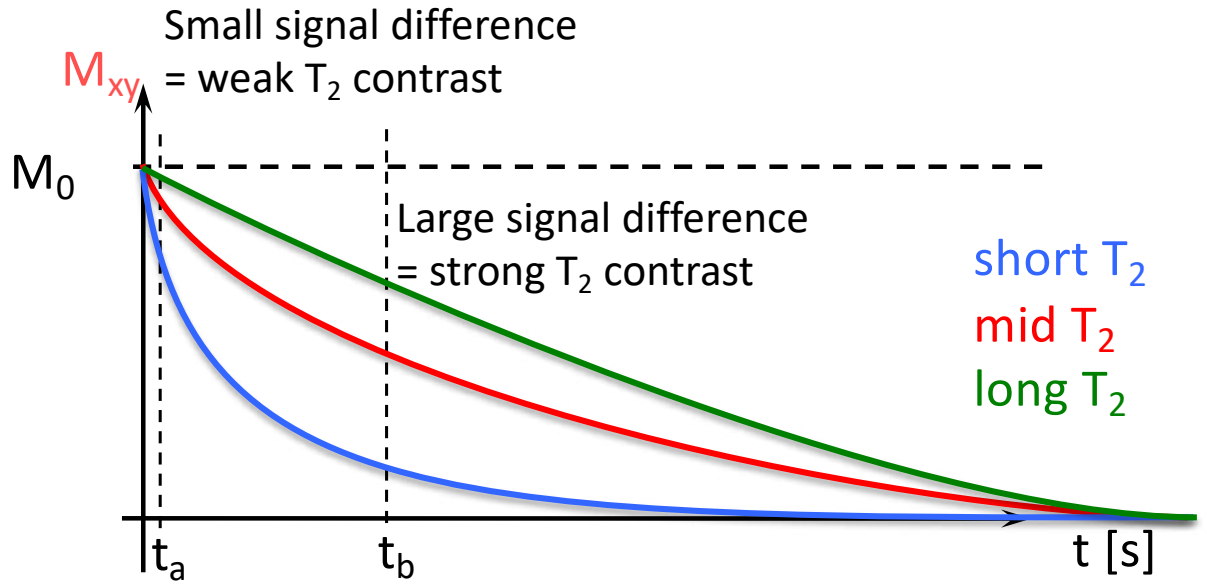
T_1 contrast

- Different tissues have different T_1 relaxation times
- T_1 contrast depend on sequence timing
  - Time after excitation (how much signal is *recovered*)



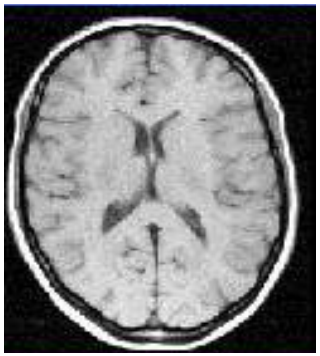
T_2 contrast

- Different tissues have different T_2 relaxation times
- T_2 contrast depend on sequence timing
  - Time after excitation (how much signal has *dephased*)

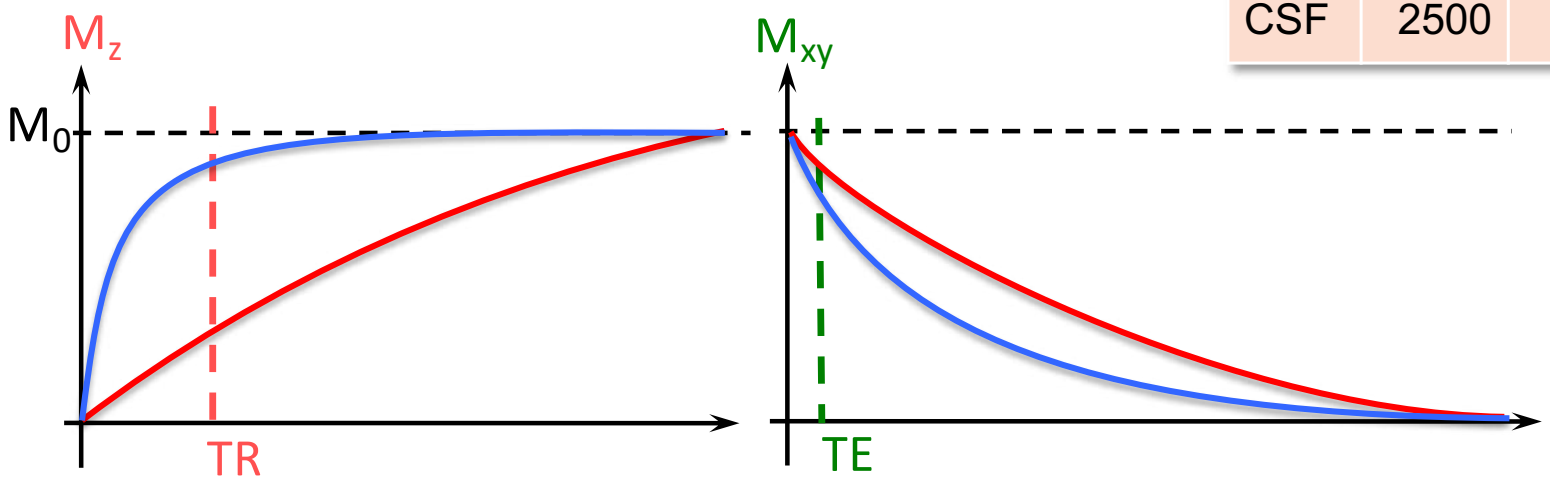


T_1 weighting

- Short TR: maximizes T_1 contrast
- Short TE: minimizes T_2 contrast
- TE cannot be made infinitely short

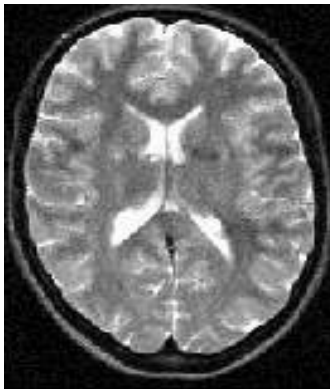


	T_1	T_2
WM	780	90
GM	920	100
CSF	2500	1000

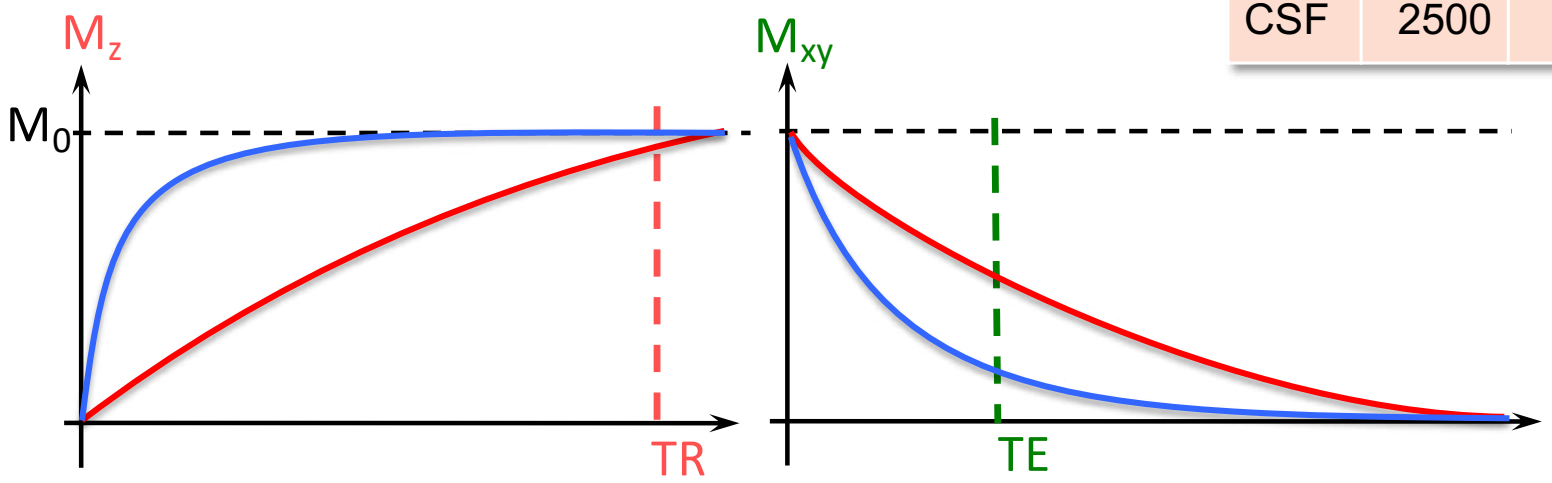


T_2 weighting

- Long TR: minimizes T_1 contrast
- Long TE: maximizes T_2 contrast
- Too long TE and all signal is lost



	T_1	T_2
WM	780	90
GM	920	100
CSF	2500	1000

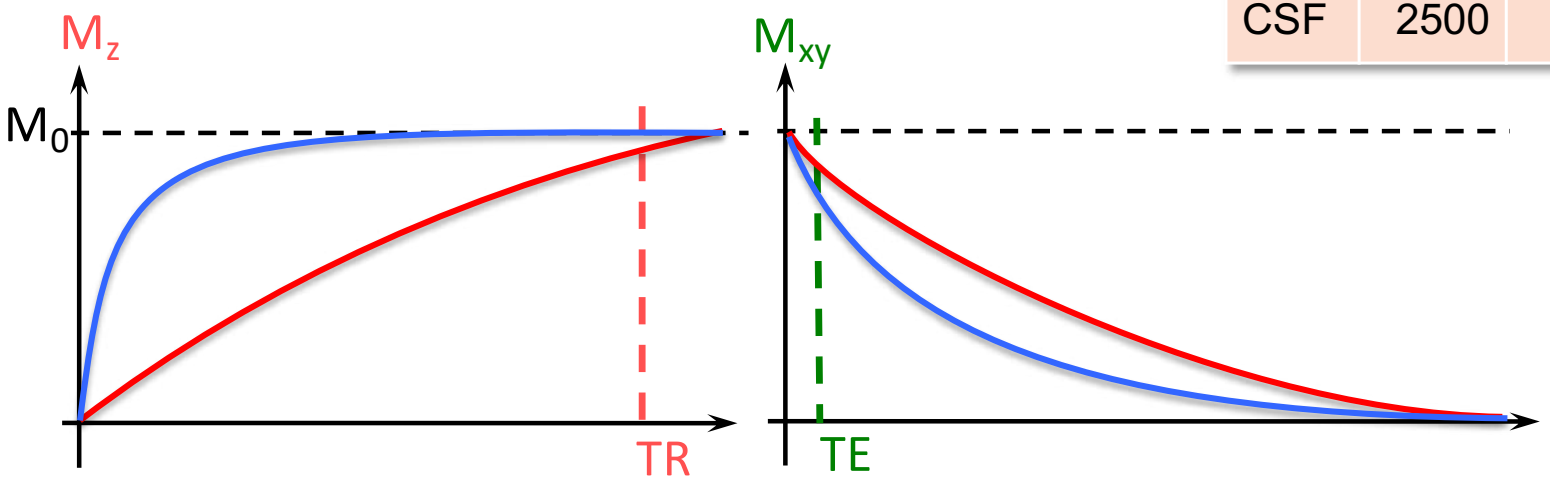


Proton Density (PD) weighting

- Long TR: minimizes T_1 contrast
- Short TE: minimizes T_2 contrast
- Remaining contrast depends on proton density

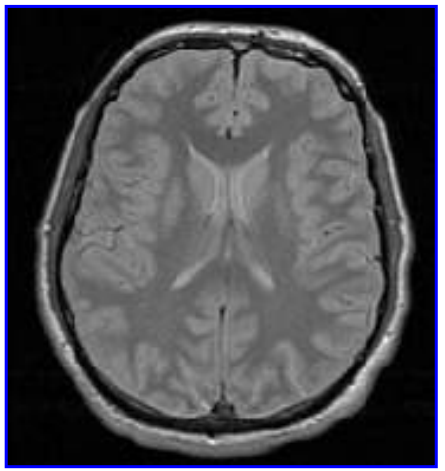


	T_1	T_2
WM	780	90
GM	920	100
CSF	2500	1000



Proton Density (PD)

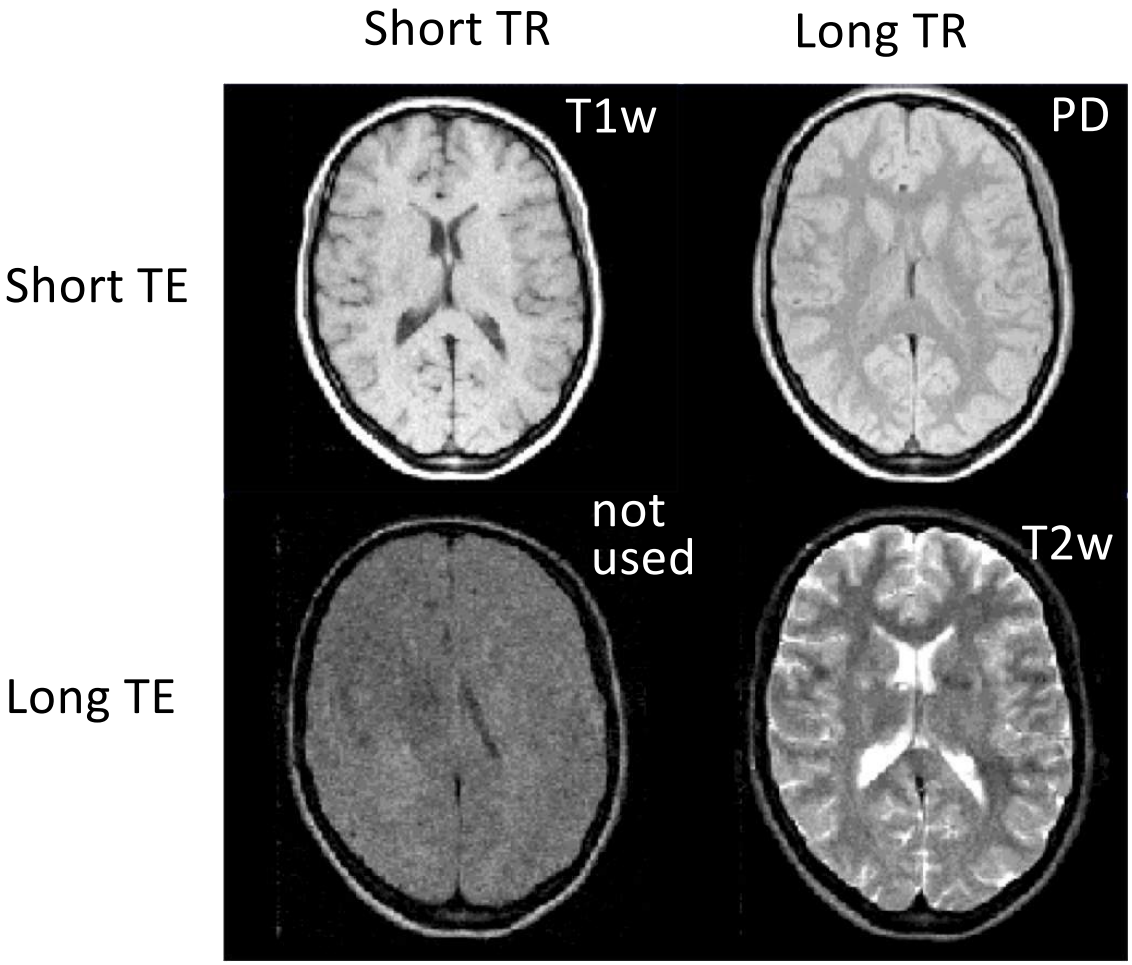
- The signal magnitude depends on the number of hydrogen nuclei (protons) in the tissue
  - -> Proton density weighting
- Different tissues have different proton densities
  - fluids: ~95 %
  - fat and water-based tissues 65-85 %
- Generally weak contrast images



Peter Jezzard

Contrast weightings

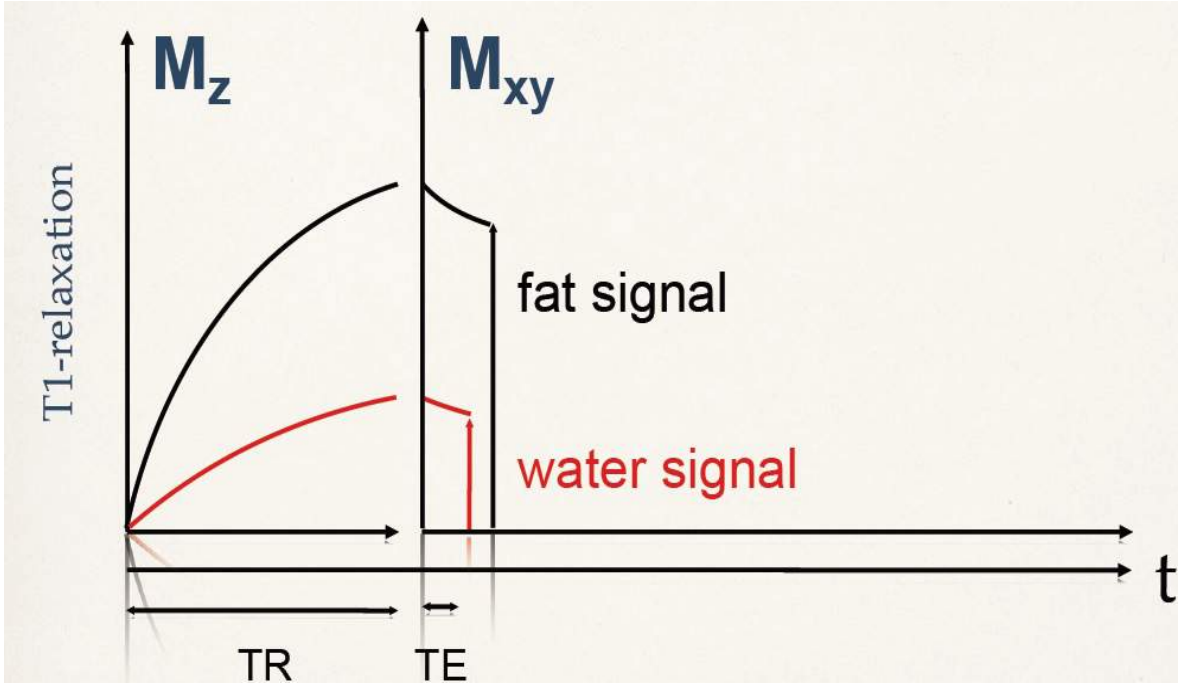
	T_1	T_2
WM	780	90
GM	920	100
CSF	2500	1000



Contrast weighting quiz 1/3

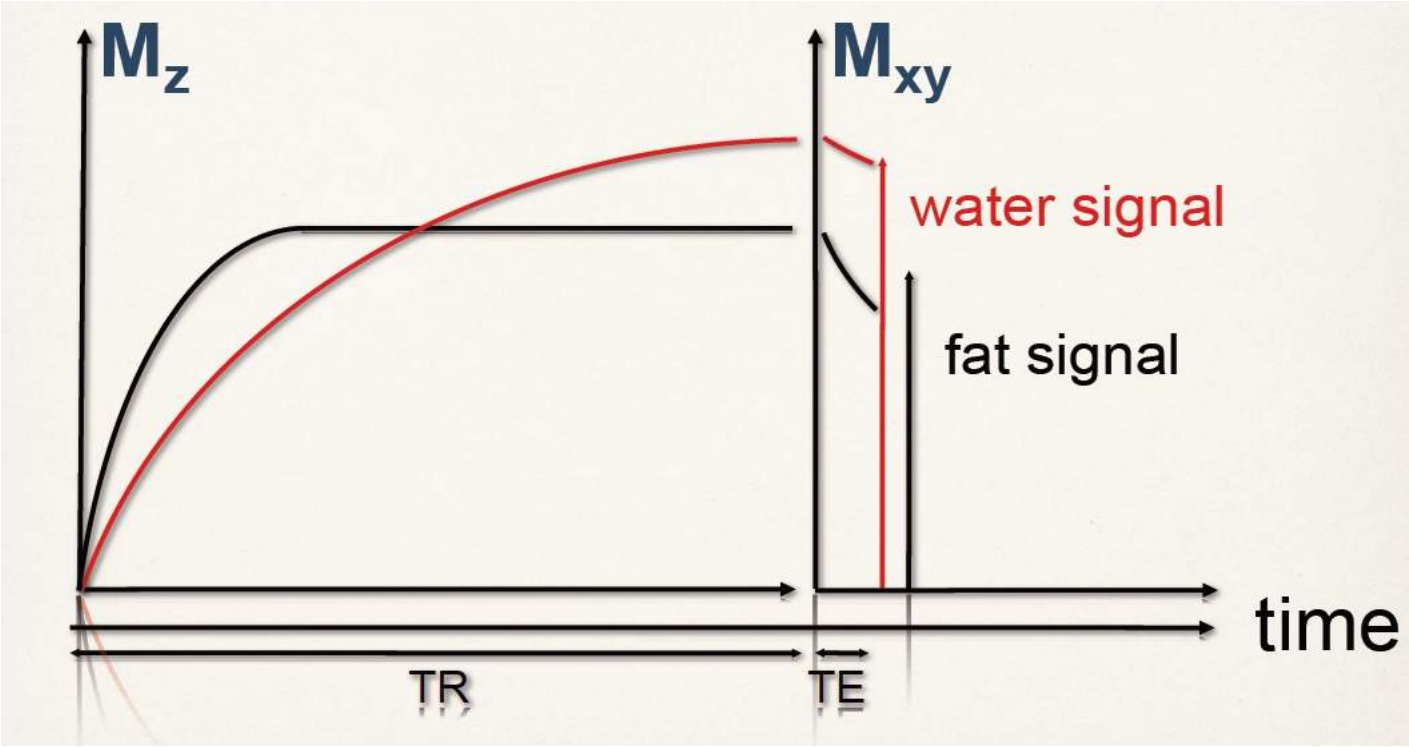
- **TE = <15 ms (short), TR = 400-600 ms (short)**

- **Water dark, fat bright - T1-weighted**



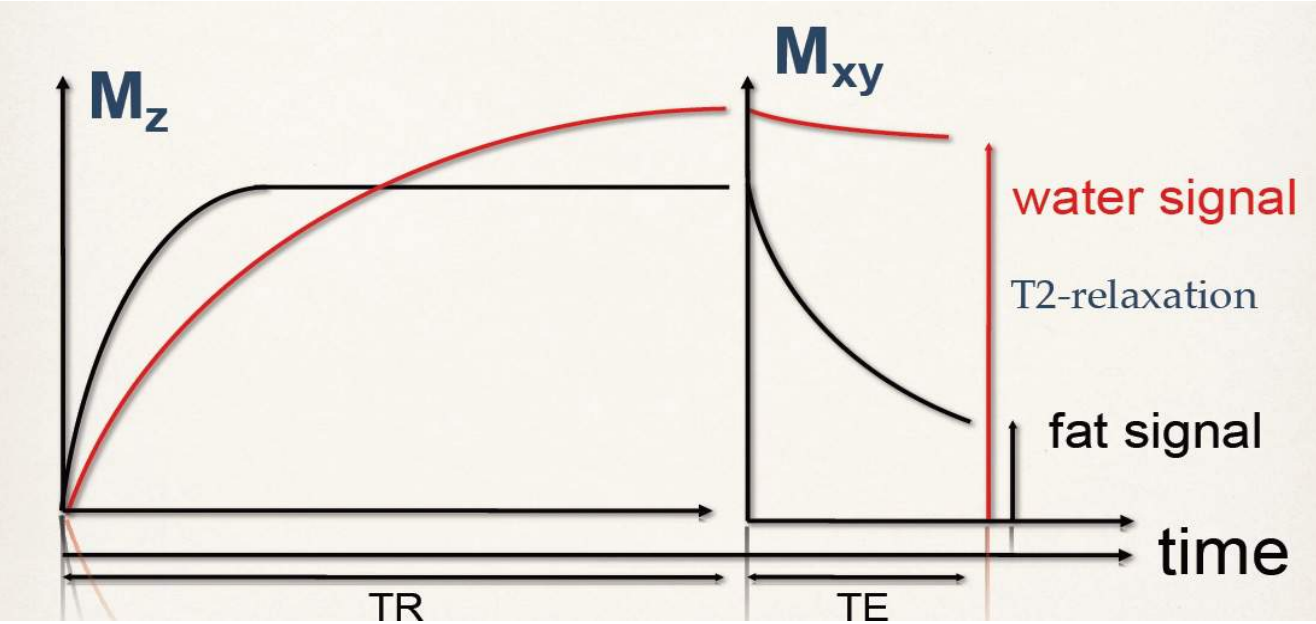
Contrast weighting quiz 2/3

- TE = <15 ms (short), TR > 4000 ms (long)
- Proton density weighted



Contrast weighting quiz 3/3

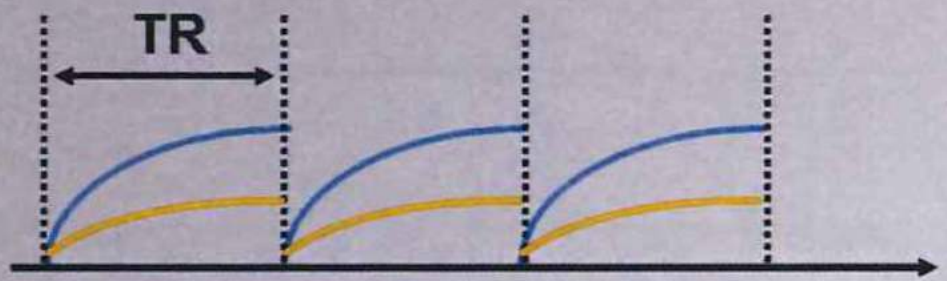
- TE = 80-90 ms (long), TR > 4000 ms (long)
- Water bright, fat dark – T_2 weighted

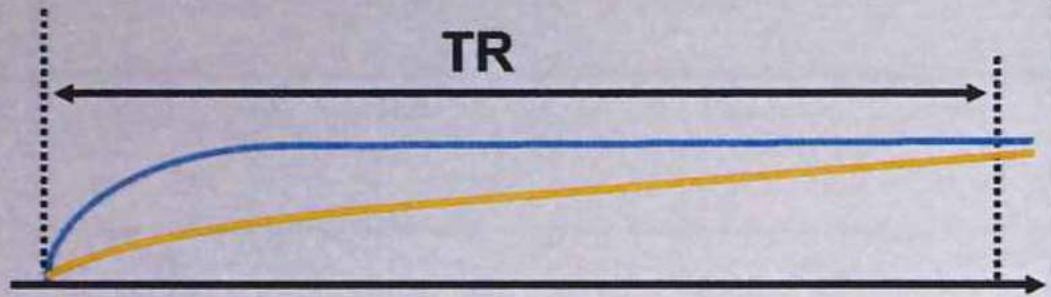


Olika TR

Effekten av lång och kort TR

SIEMENS









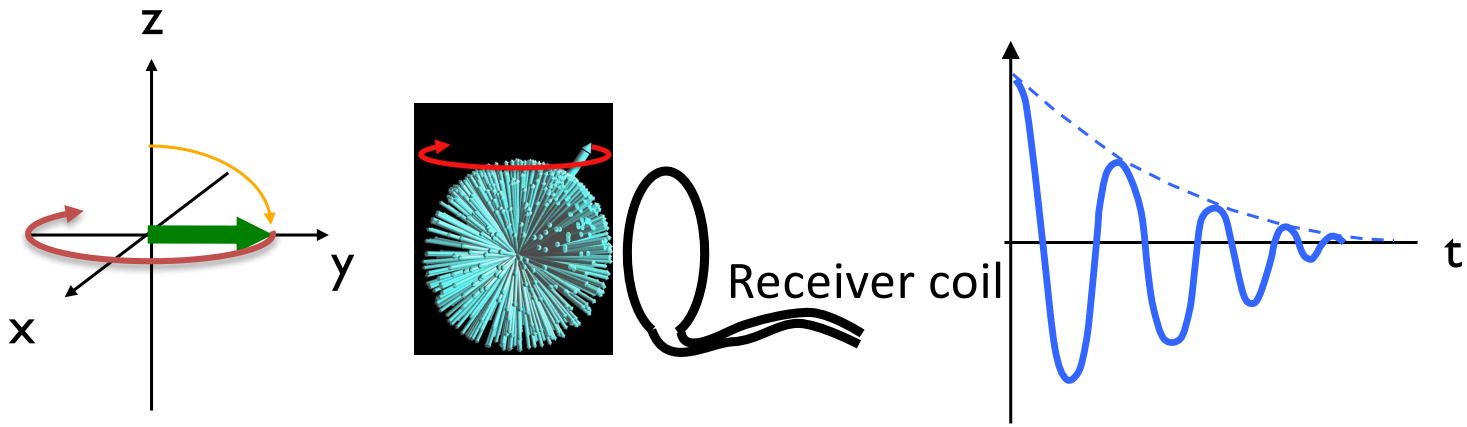
Kort TR trycker ner magnetiseringen från vävnader med lång T1 så att vävnader med kort T1 får starkare signal relativt sätt

Lång TR gör att magnetiseringen från alla vävnader hinner återhämta sig

Summary, nuclear magnetic resonance

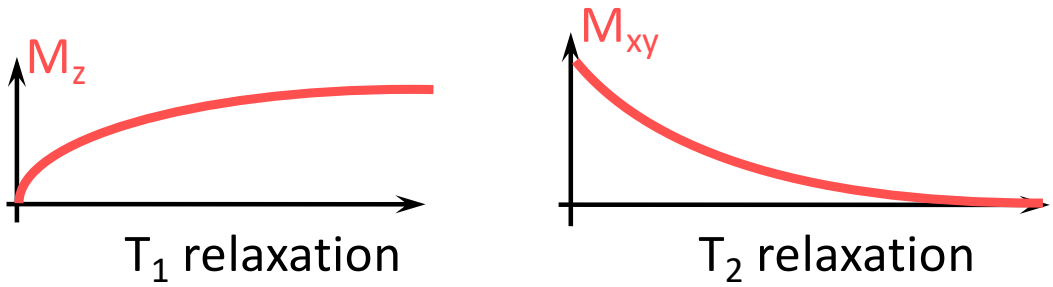


- Spins precess with a Larmor frequency proportional to external field
- Net magnetization vector can be flipped into the transverse plane M_{xy}
- The precessing net magnetization vector in M_{xy} induces current in a loop

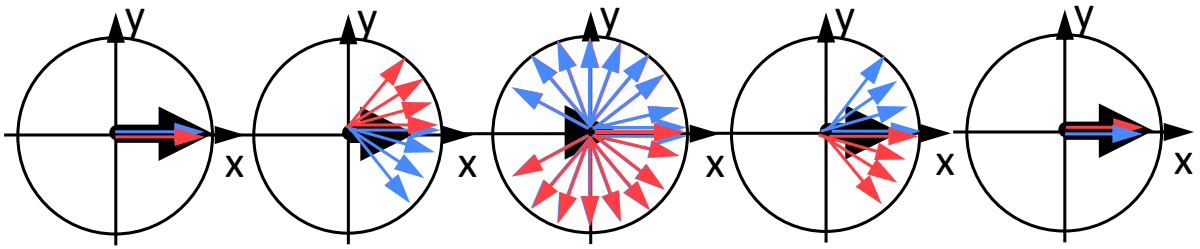


Summary, relaxation

- The longitudinal magnetization M_z returns to equilibrium; T_1 relaxation
- The transverse component dephases due to inhomogeneities; T2 relaxation



- Using spin-echo, a 180 degree refocusing pulse can recover static inhomogeneities



Summary, image contrast

- By choosing TE and TR appropriately, a desired image contrast can be obtained

